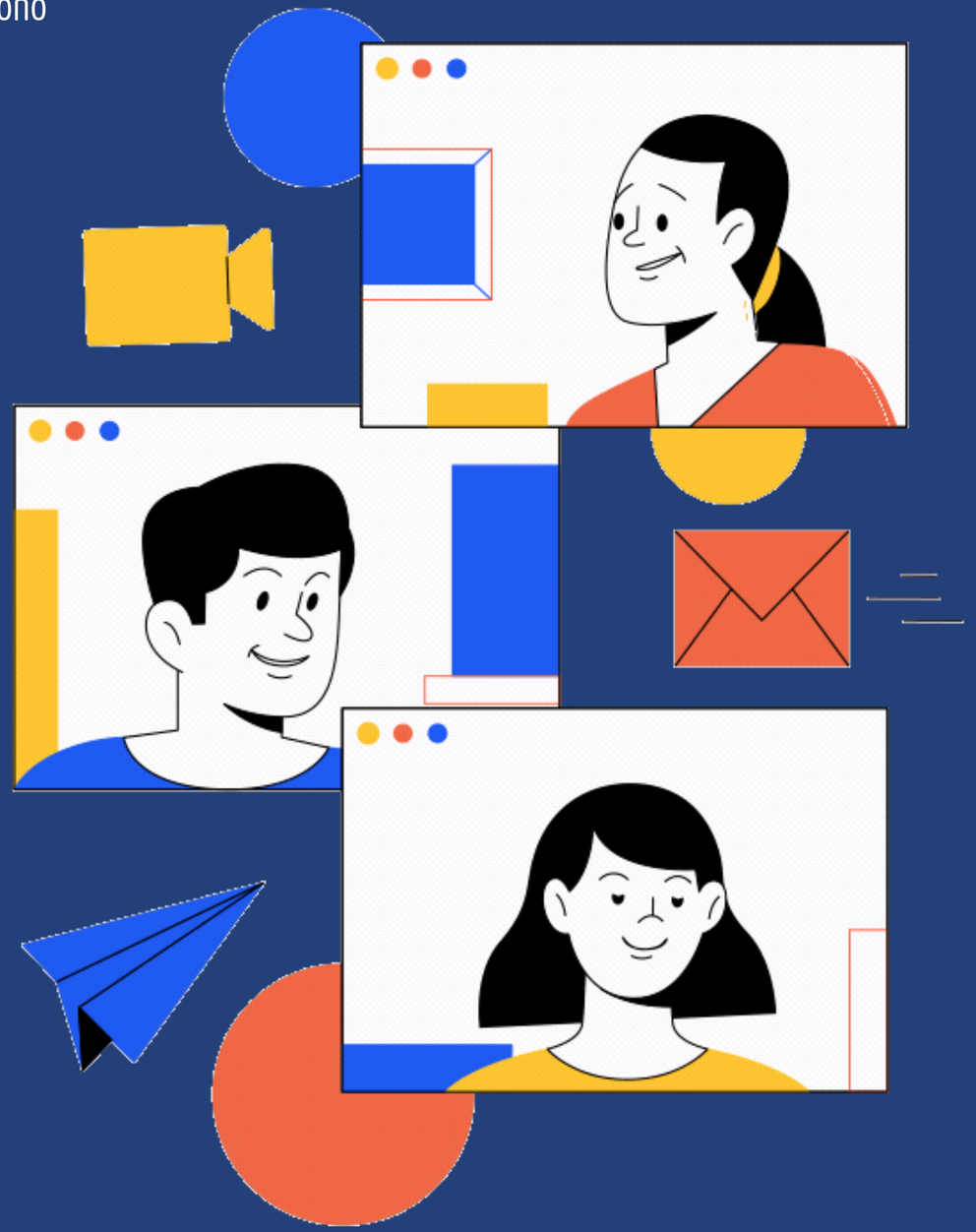
Presented by Mafud Satrio Setiono

RETAIL SALES ANALYTICS

Analyzing Sales and Customer Behavior



MAFUD SATRIO SETIONO

Data Analyst | Data Science Enthusiast

Experience

- SCM Data Analyst Intern PT Citramegah Karunia Bersama
- Data Analyst Intern Smart Hotel Kotamobagu
- Front-End Web Developer Intern PT AccSEC Priority

Data Analysis

- Python (Pandas, NumPy)
- SQL
- Excel

Data Visualization

- Power BI
- Tableau

Focus

- Turning raw data into actionable insights
- Building interactive dashboards for decision making
- Developing predictive models to support business impact

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Github

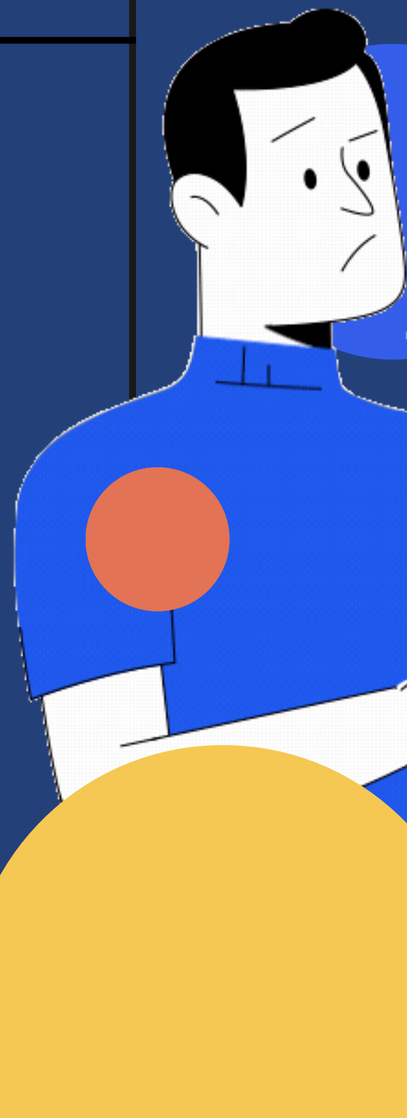
Linkedin

BUSINESS PROBLEM

Retail companies generate thousands of transactions daily, but without proper analysis it becomes difficult to identify which products, regions, and customer segments truly drive revenue.

- which products drive revenue
- which regions perform best
- how customers behave and spend

Retail managers need insights from sales and customer data to identify performance trends and improve decision making.



CORE OBJECTIVES

DATA-DRIVEN INSIGHTS ENABLE BETTER BUSINESS DECISIONS.

- Evaluate overall retail sales performance.
- Identify top performing products and regions.
- Analyze revenue contribution across sales channels.
- Segment customers using RFM analysis to understand customer value.
- Develop an interactive dashboard for business monitoring

This project aims to analyze retail sales data and customer behavior to uncover actionable insights that support business decisions.

DATA OVERVIEW

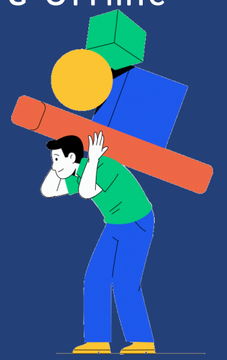
Dataset	Description
Vendas.csv	Transaction-level sales data including product, customer, store, and sales channel
Produtos.csv	Product information including category, price, and acquisition cost
Clientes.csv	Customer demographic information such as age, gender, and city
Lojas.csv	Store information including region, city, and store type

Key Features

- Customer demographics (age, gender, city)
- Product information (category, price, cost)
- Store information (region, store type)
- Sales channel (online vs offline)

Dataset Summary

- Total Transactions : 20K
- Customers : 1K
- Regions: Alentjo, Algarve, Centro
- Sales Channels : Online & Offline



Transaction-level dataset covering sales activity across multiple stores.

DATA PREPARATION

Data Quality Check

- Checked for missing values --> No missing values found
- Checked for duplicate records --> No duplicate entries detected

Data Transformation

- Converted sale_date into datetime format
- Extracted month and year for time-based analysis

Feature Engineering

- Created Revenue = quantity × unit_price
- Calculated Profit and Profit Margin
- Aggregated transaction data for dashboard metrics

Customer Dataset Preparation

- Aggregated customer transactions
- Created RFM variables:
  - Recency
  - Frequency
  - Monetary

DATA LIMITATIONS

Limitations

- The dataset only includes product acquisition cost and does not account for other operational costs such as logistics, marketing, rent, or employee salaries.
- As a result, the calculated profit margin appears unusually high compared to real-world retail businesses.
- The dataset appears to be synthetic and may not fully reflect real retail business operations.
- Product descriptions and categories originate from Portuguese retail data, which may limit contextual understanding of specific product types.

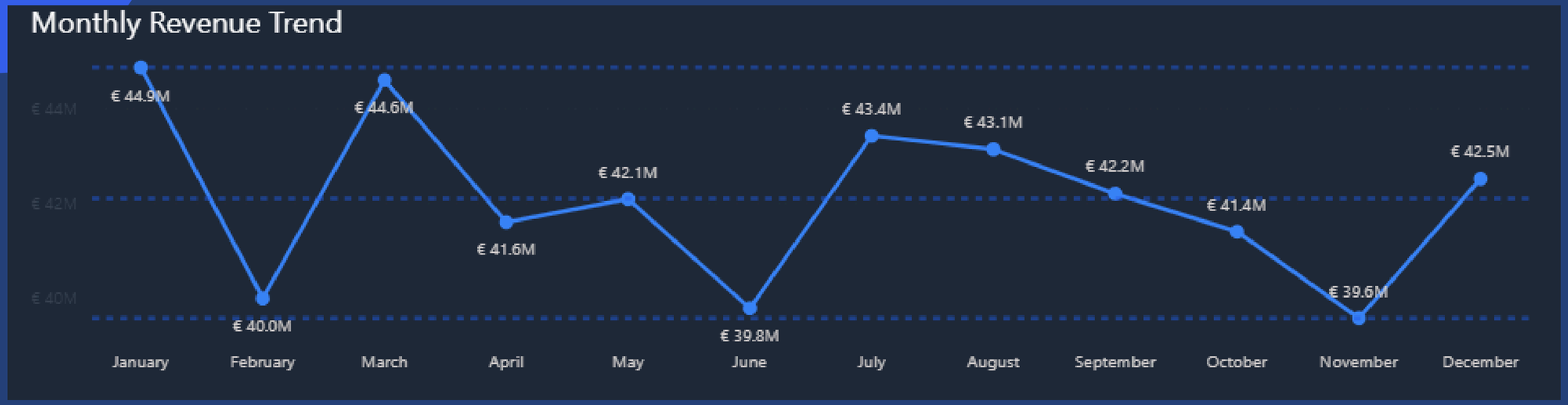
Dataset:

Retail Business Intelligence Dataset

Vendas de lojas de roupa fictícias e campanhas de marketing

k kaggle.com

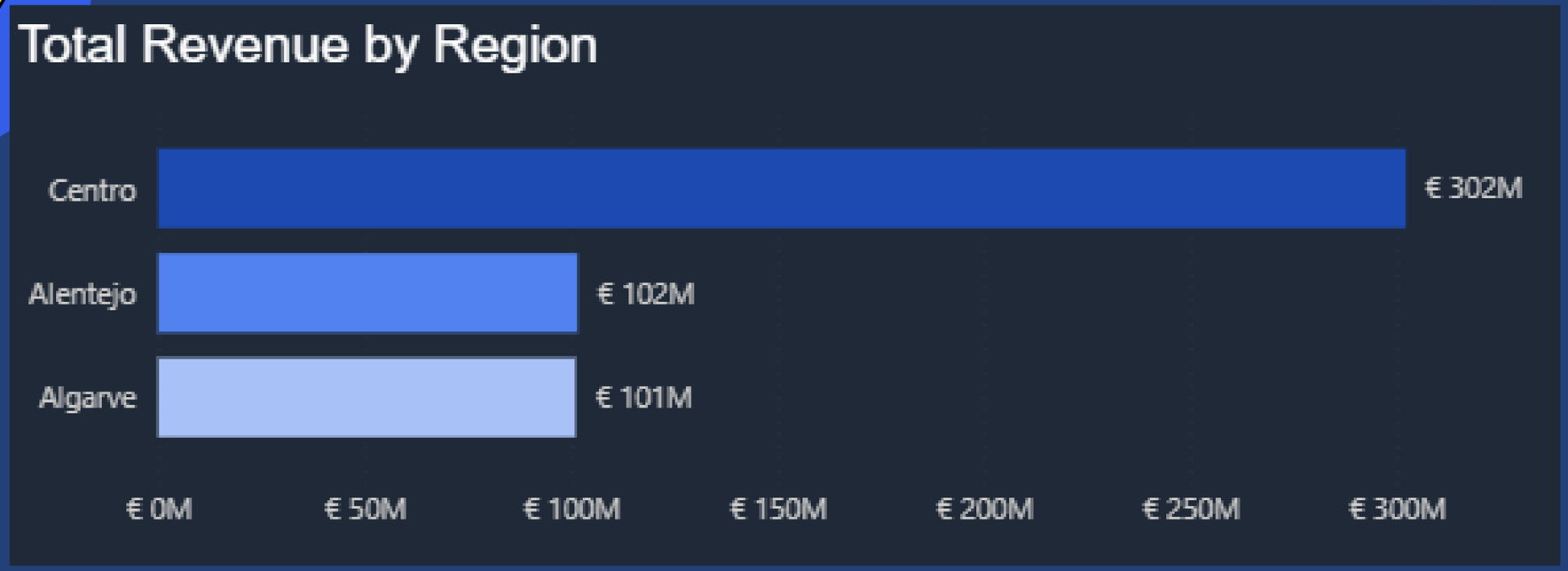
SALES TREND OVER TIME



Key Insight

- Revenue remains relatively stable throughout the year, fluctuating within a narrow range of approximately €39M to €45M, indicating consistent sales performance.
- January and March generate the highest revenue, suggesting potential seasonal demand or strong post-holiday purchasing activity.
- June and November record the lowest sales levels, indicating possible seasonal slow periods.
- Sales performance recovers toward the end of the year, with December showing a noticeable rebound.

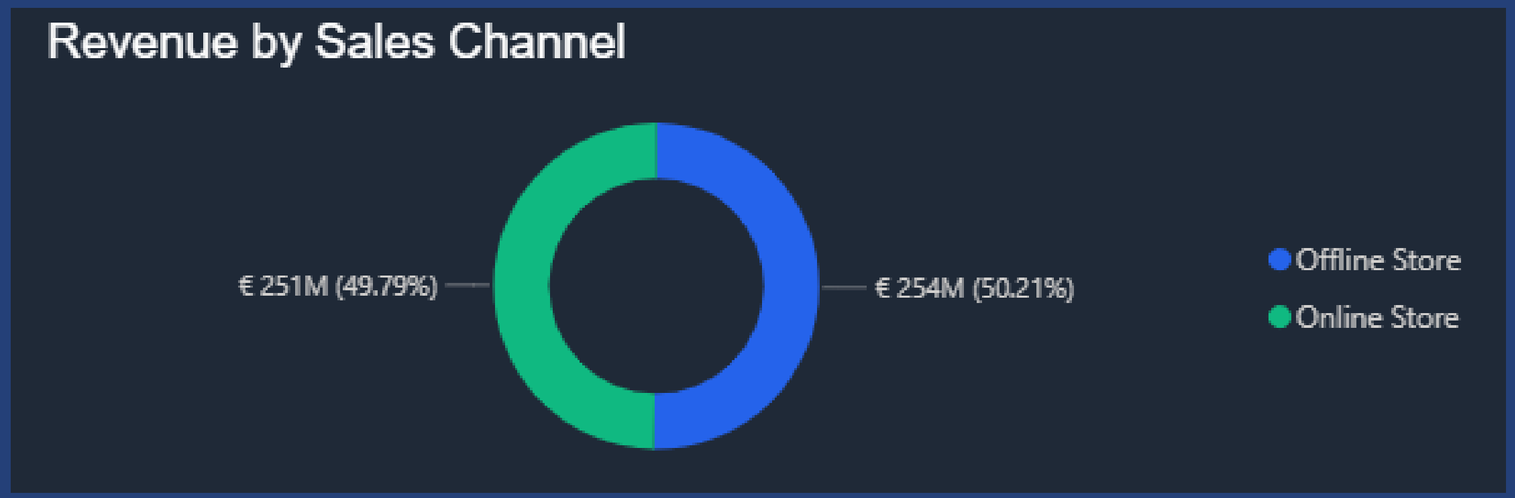
REGIONAL SALES PERFORMANCE



Key Insight

- Centro dominates total revenue, contributing approximately €302M, significantly higher than other regions.
- Alentejo and Algarve generate similar revenue levels, each contributing around €100M, indicating comparable market performance.
- The significant gap suggests that sales activity is heavily concentrated in the Centro region.

REVENUE BY SALES CHANNEL



Key Insight

- Revenue distribution between offline and online channels is nearly balanced, with offline contributing 50.21% (€254M) and online 49.79% (€251M).
- The minimal difference indicates that both sales channels play an equally important role in overall revenue generation.
- The strong performance of the online channel suggests that the business has successfully adopted an omnichannel sales strategy.

TOP 5 PRODUCTS

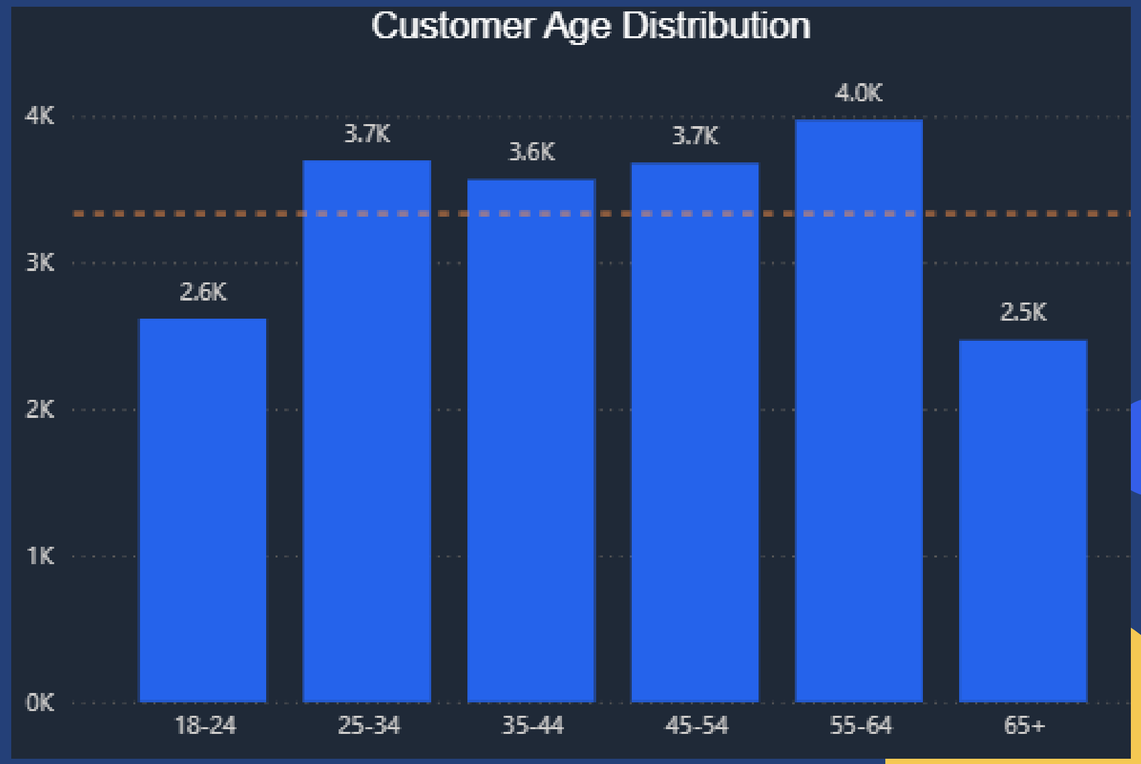
Key Insight

- Modi is the top-performing product, generating approximately €8.0M in revenue, making it the strongest revenue contributor among the top products.
- The remaining products (Adipisci, Hic, Id, and Enim) generate between €5M-€7M, indicating relatively balanced product performance.
- The gradual decline across the ranking suggests that revenue is distributed across multiple products rather than relying on a single product driver.

CUSTOMER AGE DISTRIBUTION

Key Insight

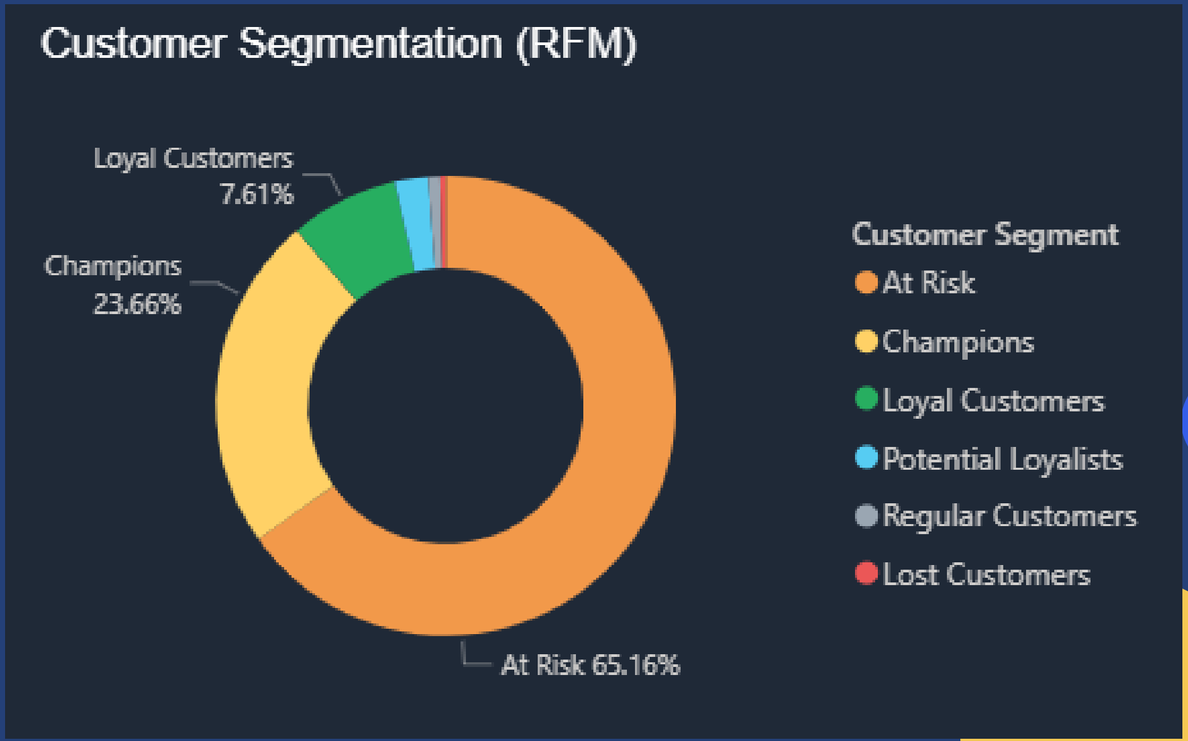
- The 55-64 age group represents the largest customer segment, with approximately 4.0K customers, indicating strong engagement from older consumers.
- Customers aged 25-54 form the core customer base, with relatively similar distribution across these age groups.
- Younger customers (18-24) and senior customers (65+) show lower participation, suggesting a smaller market share in these segments.



CUSTOMER SEGMENTATION (RFM)

Key Insight

- The high proportion of At Risk customers (65%) indicates a potential retention challenge that could impact future revenue.
- Champions represent 23.66% of the customer base, showing a valuable group of highly engaged and loyal customers.
- The relatively small share of Loyal Customers (7.61%) suggests opportunities to strengthen long-term customer relationships.



CUSTOMER SEGMENTATION (RFM)

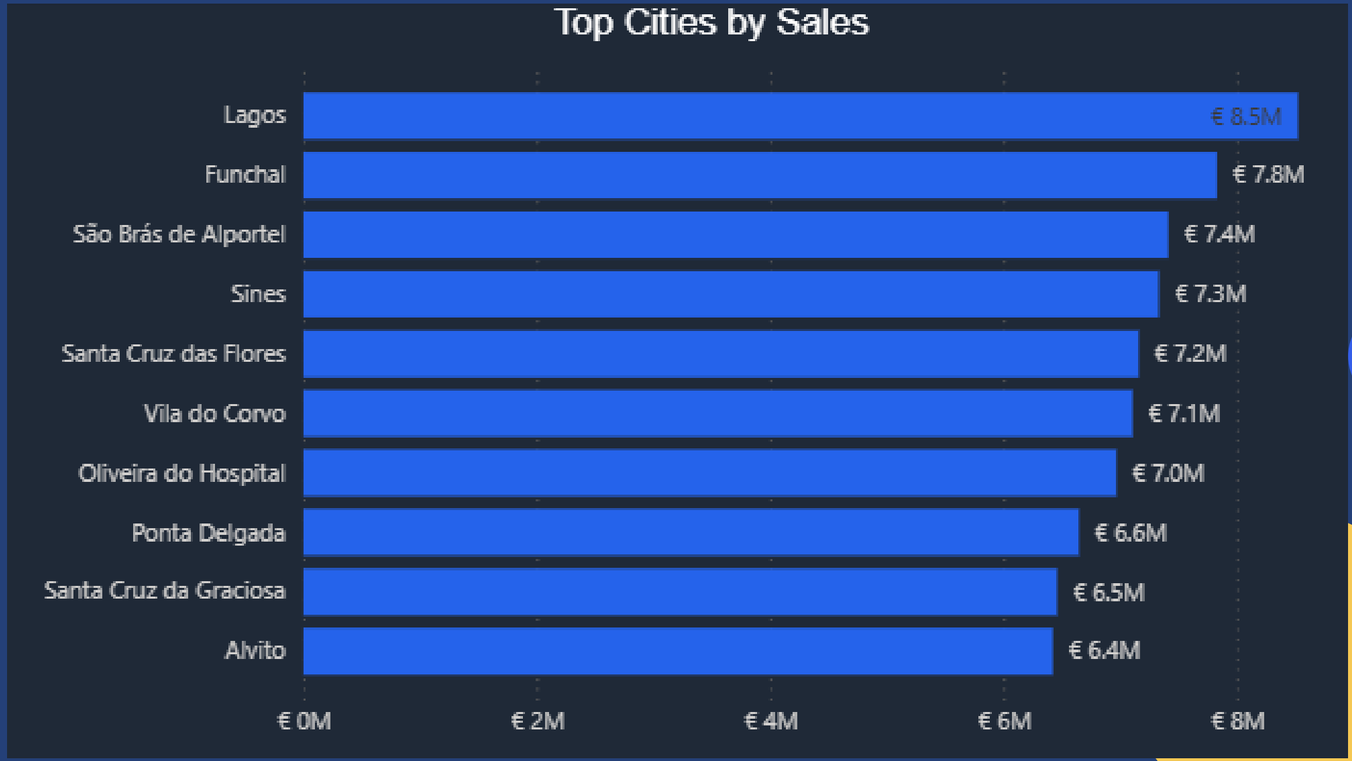
Key Insight

- Benedita Amaral is the highest-revenue customer, generating approximately €1.61M in total purchases.
- The top 10 customers each contribute over €1M in revenue, indicating a group of highly valuable customers.
- Revenue among the top customers is relatively evenly distributed, suggesting the business does not rely on a single dominant customer.

TOP CITIES BY SALES

Key Insight

- Lagos generates the highest revenue, reaching approximately €8.5M, making it the top-performing city.
- The remaining cities generate between €6.4M and €7.8M, indicating relatively balanced sales distribution across locations.
- The small revenue gap between cities suggests that sales performance is distributed across multiple urban markets rather than concentrated in a single city.



KEY BUSINESS INSIGHTS

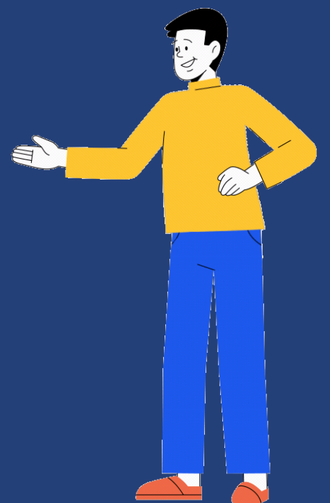
Stable Sales Performance
Revenue remains consistent throughout the year, ranging between €39M-€45M.

Centro Dominates Regional Sales
Centro contributes the largest share of revenue among all regions.

Balanced Omnichannel Strategy
Online and offline channels generate nearly equal revenue.

High Customer Retention Risk
65% of customers fall into the At-Risk segment.

Revenue Distributed Across Products & Cities
Sales performance is spread across multiple products and locations.



BUSINESS RECOMMENDATIONS

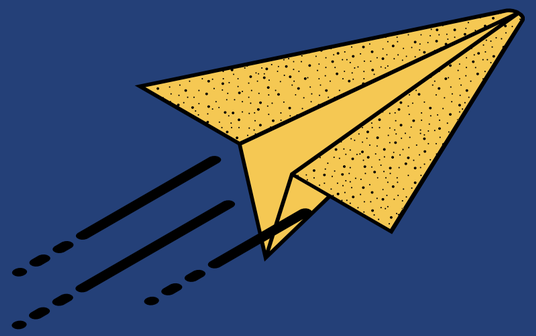
Strengthen Customer Retention Strategies
Develop targeted campaigns such as loyalty programs, personalized promotions, or re-engagement offers to reduce the high proportion of at-risk customers (65%).

Leverage High-Performing Regions
Analyze the success factors behind the Centro region's strong performance and replicate effective strategies in other regions to improve overall sales distribution.

Optimize Omnichannel Sales Strategy
Since online and offline channels contribute almost equally, businesses should continue investing in integrated omnichannel experiences to enhance customer convenience..

Promote Top-Performing Products
Increase marketing support and ensure inventory availability for high-revenue products to maximize their contribution to total sales.

Expand Market Opportunities Across Cities
Identify growth potential in cities with moderate sales performance and implement targeted marketing or store expansion strategies.



THANK YOU!